An Instantaneous Euclidean Matching Index for Multi-Carrier CCHP Capacity Planning

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***Abstract*—Capacity planning of distributed combined cooling, heating and power (CCHP) systems is usually driven by annual cost, while source–load coordination remains implicit. Conventional matching surrogates first compress each carrier over time and then aggregate the resulting scalars, obscuring simultaneous mismatch of electricity, heat, and cold. We propose the *Instantaneous Euclidean Matching Index* (IEMI), which treats the three net imbalances at each time step as a vector and uses its Euclidean norm as the magnitude of instantaneous multi-carrier mismatch. Coupled with annualized cost in a bi-objective NSGA-II framework, IEMI is compared with cost-only planning and a standard-deviation baseline on a German heating-dominated community benchmark. IEMI widens the reachable Pareto cost span from EUR 209k to EUR 1,267k. Under the same +100% budget allowance relative to the economic optimum, the selected IEMI design reduces peak grid import by 26.7% and annual grid purchase by 29.9%, while increasing self-sufficiency from 38.6% to 57.0%. The gain appears mainly in the medium-to-high investment region rather than as uniform dominance at all budgets. The L_2 choice is used to represent vector magnitude, not to claim a universally harsher penalty than L_1 aggregation.**

***Index Terms*—Integrated energy system, CCHP, source–load matching, multi-objective optimization, NSGA-II, Euclidean distance**

I. Introduction

Distributed combined cooling, heating and power (CCHP) links electricity, heat, and cold in one investment decision, so installed capacities determine for years how tightly local supply can track time-varying demand [1], [2]. Multi-objective planning often pairs annualized cost with emissions or primary-energy indicators [3], [4], yet explicit *source–load matching* is still less common as a named planning objective even though mismatch directly drives grid import, renewable spillage, and under-utilized assets [5].

Existing matching metrics often follow a two-stage logic: reduce each carrier's net-load trajectory to a scalar and then add those scalars. This loses the fact that electricity, heating, and cooling may be *simultaneously* imbalanced at hour t. Source–load coordination in CCHP is therefore a *multi-carrier time-series* problem, not only a sum of carrier-wise volatility summaries.

This paper is a **concept validation**. We introduce the **Instantaneous Euclidean Matching Index (IEMI)**: at each time step, net imbalances in electricity, heat, and cold form a three-vector, whose Euclidean length is the instantaneous multi-carrier mismatch; IEMI is the time average of that length. We compare IEMI with (i) economic-only planning and (ii) a standard-deviation baseline (Std) on one documented German community case. The emphasis is methodological clarity and a closed planning-to-operation evaluation chain.

Contributions. (1) A vector-based matching formulation that preserves instantaneous cross-carrier coupling. (2) Evidence on the German case that IEMI widens the available cost–matching design space and improves full-year operational indicators once higher budget levels are accessible. (3) A concise clarification that the L_2 norm is chosen for geometric meaning rather than as a claim of uniformly stronger penalization than L_1.

The remainder of the paper introduces the system and dispatch model, defines IEMI, presents the planning formulation and evaluation protocol, and then reports results on the German benchmark.

II. System model

We model a four-bus CCHP system with electricity, heating, cooling, and natural gas. Equipment includes PV, wind, gas turbine (CHP), electric heat pump, electric chiller, absorption chiller, and electric, thermal, and cold storages [6], [7]. Component relations are linear with fixed efficiencies/COPs, so a candidate capacity vector is evaluated through a 24-hour economic dispatch subject to bus balances and bounds.

Annualized planning cost combines investment annuities, a demand charge on peak grid import, and operating cost. To

keep the evolutionary inner loop tractable, the year is represented by 14 weighted typical days obtained from clustered load-weather profiles [8], [9]. Annual operating cost and the matching index are weighted sums over those days, while full-year 8760-hour simulation is used only after optimization to assess selected Pareto points.

III. Matching metric: IEMI

A. Why a vector at each instant

Let $P_{\text{net},i}(t)$ denote the net imbalance of carrier $i \in \{e, h, c\}$ at hour t (supply minus demand, including storage, in kW). Std-style indices use $\sigma(P_{\text{net},e}) + \sigma(P_{\text{net},h}) + \sigma(P_{\text{net},c})$ and therefore forget *when* deficits occur relative to other carriers. IEMI instead measures the *combined* mismatch at each t.

B. Instantaneous Euclidean Matching Index (IEMI)

Define the absolute mismatch vector

$$\mathbf{p}(t) = \big(|P_{\text{net},e}(t)|,\, |P_{\text{net},h}(t)|,\, |P_{\text{net},c}(t)|\big)^{\top}. \tag{1}$$

To place different carriers on comparable scales, we use fixed non-negative weights collected in $\mathbf{w} = (w_e, w_h, w_c)^{\top}$, with values determined by the case operating temperatures. The **instantaneous Euclidean mismatch** is $\big\|\mathbf{w} \odot \mathbf{p}(t)\big\|_2$, and

$$\text{IEMI} = \frac{1}{T}\sum_{t=1}^{T}\big\|\mathbf{w} \odot \mathbf{p}(t)\big\|_2. \tag{2}$$

For the German reference case, $w_e = 1$, $w_h = 0.131$, and $w_c = 0.064$. The focus of this note is the L_2 **geometry over** (e, h, c) **at fixed** t, not weight tuning; keeping w_i fixed isolates temporal-vector aggregation from carrier rescaling.

C. Euclidean (L_2) versus linear (L_1) aggregation

An alternative instantaneous index is the Manhattan magnitude

$$\text{MI}_{L1} = \frac{1}{T}\sum_{t=1}^{T}\big\|\mathbf{w} \odot \mathbf{p}(t)\big\|_1. \tag{3}$$

Both (2) and (3) are additive over time and vanish only at perfect balance. We adopt L_2 because it is the Euclidean length of the weighted mismatch vector in $\mathbb{R}^3$, i.e. a scalar measure of *simultaneous* deviation from the origin. This is a geometric choice, not a claim that L_2 always penalizes more strongly than L_1.

D. Baselines

Std (volatility baseline): $\text{MI}_{\text{Std}} = \sigma(P_{\text{net},e}) + \sigma(P_{\text{net},h}) + \sigma(P_{\text{net},c})$. **Economic**: minimize annualized cost only.

IV. Bi-objective planning framework

The three compared planning formulations share the same device set, decision variables, tariffs, and typical-day representation; only the objective set changes. The capacity vector contains rated powers of PV, WT, GT, heat pump, EC, AC, ES, HS, and CS.

Objectives. (1) Annualized total cost f_1: investment annuities, capacity charge on peak grid import, and typical-day operating cost. (2) f_2: matching index—either IEMI (2) or Std, depending on the run; Economic omits f_2.

Solver. Bi-objective problems use NSGA-II [4] with $N_{\text{ind}} = 50$ and $G_{\max} = 100$; economic-only planning uses a DE-style genetic algorithm with the same budget. Fitness evaluation parallelizes individuals, and each individual solves 14 linear programs. The IEMI run is warm-started from the final Std population, but the feasible region and search budget are unchanged; warm start is used only to accelerate convergence.

V. Case study and evaluation protocol

A. German benchmark

We report only the **German heating-dominated community** used in the shared benchmark. The input data contain 8760 hourly demand and weather profiles. Tariffs and device bounds follow the documented German configuration: electricity price 0.0025 EUR/kWh, gas price 0.0286 EUR/kWh, capacity charge 114.29 EUR/(kW·a), and non-zero bounds on all nine technologies. The tariff values are archived benchmark inputs rather than current market calibration, so the conference claim is comparative and methodological rather than prescriptive.

B. Evaluation protocol

Planning uses the 14 weighted typical days described in Section II; post-optimization assessment re-dispatches selected designs on the full 8760-hour year. Unless otherwise noted, all numbers use $N_{\text{ind}} = 50$ and $G_{\max} = 100$.

Table I is computed directly from the non-dominated sets of the three compared methods. To connect planning outcomes to operation, we then evaluate budget-normalized Pareto points with full-year dispatch. Let C_{eco} be the annualized cost of the economic optimum and define a budget allowance $\beta \in \{0, 10, 30, 50, 100\}\%$ through

$$C_{\text{tar}} = (1 + \beta)C_{\text{eco}}. \tag{4}$$

For each matching-based method, we choose the Pareto solution with the smallest matching index among designs whose annualized cost does not exceed $1.05\,C_{\text{tar}}$; otherwise the nearest-cost point is used. The resulting 8760-hour indicators form the closed evaluation chain reported in Table II.

Methods compared. Economic (A), Std (B), and IEMI (C). Std is retained as the principal baseline because it uses the same three carrier net-imbalance trajectories and the same cost-plus-matching structure as IEMI, differing mainly in whether coupling is handled instantaneously or after carrier-wise compression.

VI. Results and discussion

A. Pareto front width

Because Std and IEMI are measured on different numerical scales, we compare the *reachable cost span* of their solution sets rather than raw objective values. On the German case, IEMI spans EUR 1,267k versus EUR 209k for Std, i.e. about $6.1\times$ wider. Std already saturates at moderate investment

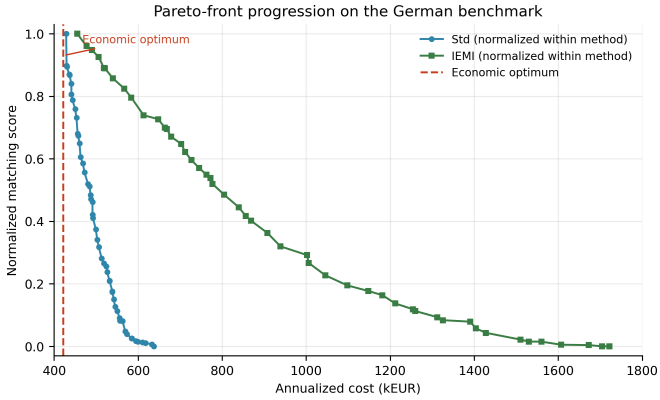

Fig. 1. Pareto-front progression after within-method normalization of the matching metric. Normalization is used only to visualize front shape and reachable cost span.

TABLE I
PARETO CHARACTERISTICS, GERMAN CASE (N_{IND}=50, G_{MAX}=100).

	A Economic	**B** Std	**C** IEMI
Pareto points stored	1	50	50
Min. cost (kEUR)	422	429	455
Max. cost (kEUR)	—	637	1,722
Front width (kEUR)	—	209	**1,267**
Ratio vs. Std	—	1.0×	6.1×

levels, whereas IEMI continues to generate distinct high-cost trade-offs.

Fig. 1 shows the same fronts after within-method normalization, avoiding the misleading impression that Std and IEMI objective values are directly comparable in magnitude.

B. Budget-aligned operational indicators

Selected Pareto points are re-simulated over 8760 hours using the protocol of Section V. Table II reports the +100% budget-allowance case. The listed annualized costs are the *actual* costs of the selected designs: Std saturates at EUR 637k, whereas IEMI still offers lower-mismatch alternatives and reaches EUR 869k within the allowed tolerance band.

Relative to Std under the same budget allowance, IEMI reduces peak grid import by 26.7%, annual grid purchase by 29.9%, and raises self-sufficiency by 18.3 percentage points. Curtailment remains negligible in both cases, reaching only 0.08% for the selected IEMI design. The wider IEMI solution set therefore translates into materially different year-round grid interaction.

Fig. 2 also shows that IEMI is *not* uniformly superior at every budget level. At +10% and +30%, Std still gives slightly lower grid purchase and slightly higher self-sufficiency. The crossover appears around +50%, after which IEMI dominates the reported grid-interaction indicators. The selected +100% IEMI design also shifts more strongly toward PV, GT, absorption cooling, and cold storage while largely eliminating electric heat-pump and electric-chiller capacity, which is consistent with its lower electric dependence.

TABLE II
OPERATIONAL INDICATORS AT +100% BUDGET, GERMAN CASE (SELECTED PARETO POINTS).

Method	Cost (kEUR)	Peak grid (kW)	Annual grid (MWh)	Self-suff. (%)
Std	637	2,617	15,374	38.6
IEMI	869	**1,920**	**10,781**	**57.0**

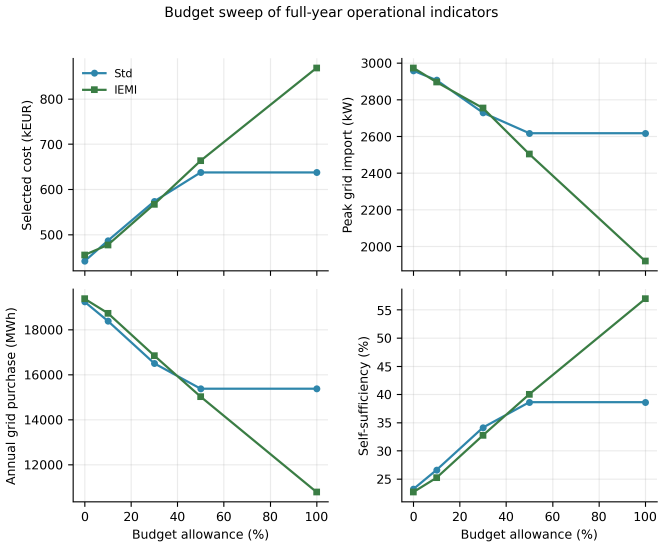

Fig. 2. Full-year operating indicators of budget-aligned designs. Std improves slightly faster under tight budgets, but saturates after +50%; IEMI overtakes once its wider high-investment region becomes accessible.

C. Structural interpretation

Std aggregates per-carrier volatility before cross-carrier coupling and becomes relatively insensitive once those fluctuations are already compressed, so NSGA-II sees little further variation in f_2. IEMI preserves hour-by-hour coupling across electricity, heat, and cooling, so designs that reduce *simultaneous* multi-carrier mismatch remain distinguishable as investment increases.

D. Scope and threats to validity

Three cautions bound the present claim. First, Std and IEMI should not be compared through raw f_2 magnitude; we therefore rely on front shape, reachable cost span, and 8760-hour downstream indicators. Second, the German case uses archived benchmark tariffs, so the absolute capacity mix is benchmark-specific. Third, the planning stage uses 14 representative days and one documented evolutionary setting; full-year re-dispatch partly mitigates temporal aggregation bias, but broader evidence across climates, tariffs, and random seeds remains future work.

VII. CONCLUSION

We introduced the Instantaneous Euclidean Matching Index (IEMI) for CCHP capacity planning: a time average of the Euclidean norm of the three-carrier net-imbalance vector. On the German community case, IEMI widens the attainable planning trade-off range relative to a standard-deviation baseline

and, under the same +100% budget allowance, improves full-year peak import, annual grid purchase, and self-sufficiency. The gain appears mainly once medium-to-high investment levels become accessible. The L_2 choice reflects geometric magnitude of joint mismatch, not a claim of universally stronger penalization than L_1.

The evidence chain is explicit: typical-day optimization generates candidate designs, budget-normalized Pareto selection chooses comparable points, and 8760-hour re-dispatch confirms the operational consequences. The present note is intentionally limited to one heating-dominated benchmark with fixed carrier weights and benchmark-specific tariffs; broader robustness and sensitivity analysis are natural next steps for a journal extension.

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